

**AMITY INSTITUTE OF INFORMATION
TECHNOLOGY
(AIIT)**



MASTER OF COMPUTER APPLICATIONS

Laboratory Record

Web Design and Development

CSE5009

2025-2026

SUBMITTED TO:

Dr. MADHU BHAN
Associate Professor

SUBMITTED BY:

NAME: HANAMANTH B KURI
SEN:A869145025045

Amity University Bengaluru

NH-207, opposite to the office of Deputy Commissioner, Devanahalli,
Doddaballapura, Bangalore – 562110

E-Learning Web Application Using HTML, CSS, and JavaScript

Abstract

The E-Learning Web Application aims to simplify online education by offering a centralized platform for course exploration and student interaction. The primary objective of this project is to design and develop a responsive, visually appealing, and easy-to-use learning platform using HTML, CSS, and JavaScript. The application includes essential features such as user registration, secure login validation, course listing, enrollment functionality, profile creation, and contact forms.

The methodology involves client-side validation, modular page design, and browser-based data storage using localStorage. Key outcomes include improved user experience, efficient course enrollment, and structured navigation across different modules. This project demonstrates how front-end web technologies can be effectively used to build functional educational platforms without server-side dependencies.

Introduction

With the rapid growth of digital learning, e-learning platforms have become essential in modern education systems. Traditional classroom learning is increasingly being supplemented by online platforms that allow learners to access content anytime and anywhere. This project addresses the need for a simple yet effective e-learning interface suitable for students and beginners.

The scope of this project is limited to front-end development but covers major functionalities required in real-world learning management systems. The project is highly relevant for understanding web development fundamentals such as UI design, form validation, navigation flow, and data handling in browsers. It serves as a strong foundation for further enhancement using backend technologies and databases.

METHODOLOGY

The development of the E-Learning Web Application follows a structured and step-by-step approach:

1. Requirement Analysis

The system requirements were identified based on common e-learning needs such as authentication, course display, enrollment, and profile management.

2. Design Phase

User interface designs were created using CSS animations, gradients, and responsive layouts to enhance user engagement. Navigation bars and modular pages ensure easy accessibility.

3. Tools and Technologies Used

HTML5 – Structure of web pages

CSS3 – Styling, animations, gradients, and responsiveness

JavaScript – Form validation, event handling, and data storage

LocalStorage – Client-side data persistence

4. Development Phase

Each module was developed separately:

Login and Registration pages

Course listing and enrollment modal

Profile creation and viewing

Contact form with validation

5. Testing

Manual testing was performed to validate form inputs, navigation flow, and storage functionality.

Flow of the System

User → Register/Login → Home Page → Courses → Enrollment → Profile → Logout

IMPLEMENTATION & RESULTS

Implementation

The project consists of multiple interconnected web pages:

Login & Registration: Validates user credentials using JavaScript.

Home Page: Acts as a dashboard with navigation options.

Courses Page: Displays multiple courses with an “Enroll Now” option.

Enrollment Modal: Captures student details and confirms enrollment.

Profile Module: Allows students to create and view personal profiles.

Contact Page: Collects user queries with input validation.

Data such as user details and selected courses are stored using **localStorage**, enabling temporary persistence without backend integration.

Results and Observations

Smooth navigation across all modules

Responsive and attractive UI

Effective form validation

Successful client-side data storage

Improved user interaction using animations and modals

The project successfully meets its objectives and demonstrates practical implementation of front-end web development concepts.

CONCLUSION & REFERENCES

Conclusion

The E-Learning Web Application project successfully demonstrates how modern web technologies can be used to create a functional and interactive learning platform. The system provides essential features such as user authentication, course enrollment, and profile management while maintaining simplicity and usability.

Challenges Faced

- Managing data without backend support

- Ensuring consistent UI across pages

- Implementing validations using JavaScript

Future Scope

- Backend integration using PHP/Node.js

- Database connectivity (MySQL/MongoDB)

- Admin dashboard

- Course progress tracking

- Online assessments and certificates

References

- 1.MDN Web Docs – HTML, CSS, JavaScript
- 2.W3Schools Web Development Tutorials
- 3.JavaScript LocalStorage Documentation
- 4.UI Design Principles for Web Applications

OUTPUT



